Assignment of Chapter Two (C)

A. **Fill-in-Blank：**

1 The simplification of the logic algebra contains three common methods, which are__________________， __________________， and __________________.

2 Each square in a Karnaugh map represents a __________________.

3 A Karnaugh map with *n* variables consists of __________________ (how many) squares.

4 “The Karnaugh map is suitable to simplify the functions with many variables.” This statement is __________________(true/false).

B. Use the algebra simplification method to find the simplest “AND-OR” expressions of the following logic functions.

1， $F = \overline{A}\,\overline{B} + \overline{B}\ \overline{C} + BC + AB$

2， $F = (A + B + C)(\overline{A} + B)(A + B + \overline{C})(C + D)$

3， $F = AB\overline{C} + \overline{A}BC + A\overline{B}\,\overline{C} + \overline{A}\,\overline{C}$

4, $F = AB + \overline{A}\,\overline{B}C + BC$

5, $F = A\overline{B} + B + BCD$

6, $F = (A+B+C)(\overline{A}+B)(A+B+\overline{C})$

C. Use the Karnaugh Map simplification method to find the simplest “AND-OR” expressions of the following logic functions.

1， $F(A,B,C)=\sum m(0,1,2,4,5,7)$

2， $F(A,B,C,D) = \prod M(1,5,6,7,11,12,13,15)$

3， $F(\mathrm{A,B,C,D}) = \sum m(0,1,2,3,4,6,7,8,9,11,15)$

D. Use the Karnaugh Map simplification method to find the simplest "AND-OR" and "OR-AND" expressions of the following logic functions.

1, $F(A,B,C,D) = \overline{A}\overline{B} + \overline{A}\overline{C}D + AC + B\overline{C}$

2, $F(A,B,C,D) = \prod M(2,4,6,10,11,12,13,14,15)$